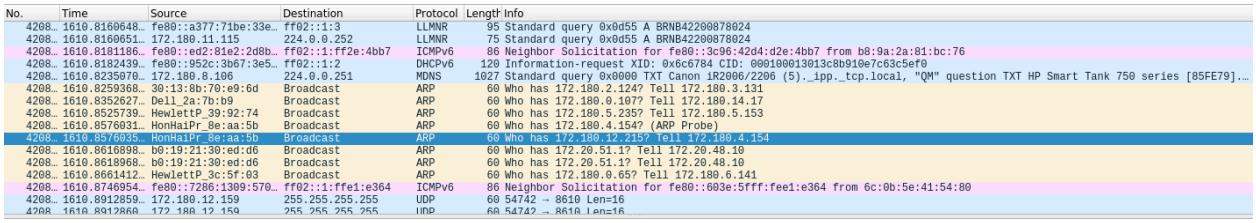
LAB TASK-7

TASK-1:

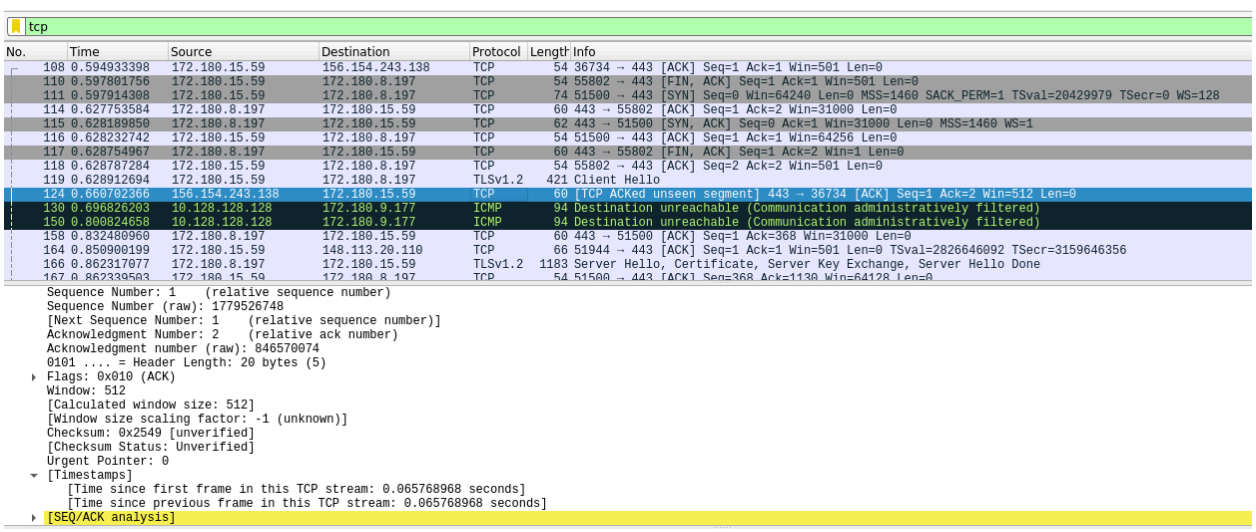
Packet capturing :



No.	Time	Source	Destination	Protocol	Length	Info
4208…	1610.8160648…	fe80::a377:71be:33e…	ff02::1:3	LLMNR	95	Standard query 0x0d55 A BRNB42200878024
4208…	1610.8160651…	172.180.11.115	224.0.0.252	LLMNR	75	Standard query 0x0d55 A BRNB42200878024
4208…	1610.8181186…	fe80::ed2:81e2:2d8b…	ff02::1:ff2e:4bb7	ICMPv6	86	Neighbor Solicitation for fe80::3c96:42d4:d2e:4bb7 from b8:9a:2a:81:bc:76
4208…	1610.8182439…	fe80::952c:3b67:3e5…	ff02::1:2	DHCPv6	120	Information-request XID: 0x6c6784 CID: 000100013013c8b910e7c63c5ef0
4208…	1610.8235070…	172.180.8.106	224.0.0.251	MDNS	1027	Standard query 0x0000 TXT Canon iR2006/2206 (5)._ipp._tcp.local, "QM" question TXT HP Smart Tank 750 series [85FE79].…
4208…	1610.8259368…	30:13:8b:70:e9:6d	Broadcast	ARP	60	Who has 172.180.2.124? Tell 172.180.3.131
4208…	1610.8352627…	Dell_2a:7b:b9	Broadcast	ARP	60	Who has 172.180.0.107? Tell 172.180.14.17
4208…	1610.8525739…	HewlettP_39:92:74	Broadcast	ARP	60	Who has 172.180.5.235? Tell 172.180.5.153
4208…	1610.8576031…	HonHaiPr_8e:aa:5b	Broadcast	ARP	60	Who has 172.180.4.154? (ARP Probe)
4208…	1610.8576035…	HonHaiPr_8e:aa:5b	Broadcast	ARP	60	Who has 172.180.12.215? Tell 172.180.4.154
4208…	1610.8616898…	b0:19:21:30:ed:d6	Broadcast	ARP	60	Who has 172.20.51.1? Tell 172.20.48.10
4208…	1610.8618968…	b0:19:21:30:ed:d6	Broadcast	ARP	60	Who has 172.20.51.1? Tell 172.20.48.10
4208…	1610.8661412…	HewlettP_3c:5f:03	Broadcast	ARP	60	Who has 172.180.0.65? Tell 172.180.6.141
4208…	1610.8746954…	fe80::7286:1309:570…	ff02::1:ff e1:e364	ICMPv6	86	Neighbor Solicitation for fe80::603e:5fff:fee1:e364 from 6c:0b:5e:41:54:80
4208…	1610.8912859…	172.180.12.159	255.255.255.255	UDP	60	54742 → 8610 Len=16
4208…	1610.8912860…	172.180.12.159	255.255.255.255	UDP	60	54742 → 8610 Len=16

Packet capturing is the process of intercepting and recording network traffic as it travels across a network, typically for analysis, troubleshooting, or security monitoring. It is performed using specialized software tools such as Wireshark or tcpdump, which place a network interface card (NIC) into promiscuous or monitor mode so it can capture all packets passing through, not just those addressed to it. The captured data includes packet headers (containing source and destination IP addresses, protocols, and ports) and sometimes payloads, depending on configuration and encryption. These packets are then stored and analyzed to understand communication patterns, detect anomalies, diagnose network issues, or investigate potential security threats.

Filtering :



tcp

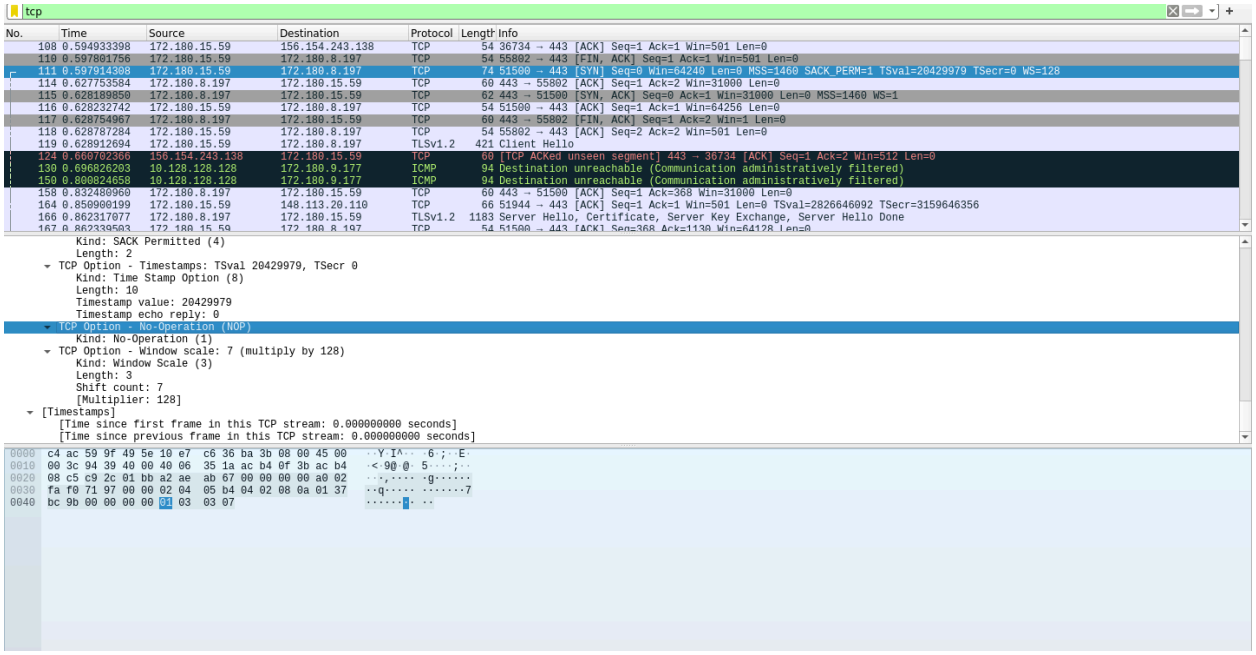
No.	Time	Source	Destination	Protocol	Length	Info
108	0.594933398	172.180.15.59	156.154.243.138	TCP	54	36734 → 443 [ACK] Seq=1 Ack=1 Win=501 Len=0
110	0.597801756	172.180.15.59	172.180.8.197	TCP	54	55802 → 443 [FIN, ACK] Seq=1 Ack=1 Win=501 Len=0
111	0.597914308	172.180.15.59	172.180.8.197	TCP	74	51500 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=20429979 TSecr=0 WS=128
114	0.627753584	172.180.8.197	172.180.15.59	TCP	60	443 → 55802 [ACK] Seq=1 Ack=2 Win=31000 Len=0
115	0.628189850	172.180.8.197	172.180.15.59	TCP	62	443 → 51500 [SYN, ACK] Seq=0 Ack=1 Win=31000 Len=0 MSS=1460 WS=1
116	0.628232742	172.180.15.59	172.180.8.197	TCP	54	51500 → 443 [ACK] Seq=1 Ack=1 Win=64256 Len=0
117	0.628754967	172.180.8.197	172.180.15.59	TCP	60	443 → 55802 [FIN, ACK] Seq=1 Ack=2 Win=1 Len=0
118	0.628787284	172.180.15.59	172.180.8.197	TCP	54	55802 → 443 [ACK] Seq=2 Ack=2 Win=501 Len=0
119	0.628912694	172.180.15.59	172.180.8.197	TLSv1.2	421	Client Hello
124	0.660702366	156.154.243.138	172.180.15.59	TCP	60	[TCP ACKed unseen segment] 443 → 36734 [ACK] Seq=1 Ack=2 Win=512 Len=0
130	0.696826203	10.128.128.128	172.180.9.177	ICMP	94	Destination unreachable (Communication administratively filtered)
150	0.800824658	10.128.128.128	172.180.9.177	ICMP	94	Destination unreachable (Communication administratively filtered)
158	0.832480960	172.180.8.197	172.180.15.59	TCP	60	443 → 51500 [ACK] Seq=1 Ack=368 Win=31000 Len=0
164	0.850900199	172.180.15.59	148.113.20.110	TCP	66	51944 → 443 [ACK] Seq=1 Ack=1 Win=501 Len=0 TSval=2826646092 TSecr=3159646356
166	0.862317077	172.180.8.197	172.180.15.59	TLSv1.2	1183	Server Hello, Certificate, Server Key Exchange, Server Hello Done
167	0.862339503	172.180.15.59	172.180.8.197	TCP	54	51500 → 443 [ACK] Seq=368 Ack=1130 Win=64128 Len=0

```
    Sequence Number: 1    (relative sequence number)
    Sequence Number (raw): 1779526748
    [Next Sequence Number: 1    (relative sequence number)]
    Acknowledgment Number: 2    (relative ack number)
    Acknowledgment number (raw): 846570074
    0101 .... = Header Length: 20 bytes (5)
  ▸ Flags: 0x010 (ACK)
    Window: 512
    [Calculated window size: 512]
    [Window size scaling factor: -1 (unknown)]
    Checksum: 0x2549 [unverified]
    [Checksum Status: Unverified]
    Urgent Pointer: 0
  ▾ [Timestamps]
      [Time since first frame in this TCP stream: 0.065768968 seconds]
      [Time since previous frame in this TCP stream: 0.065768968 seconds]
  ▸ [SEQ/ACK analysis]
```

Packet filtering is the process of analyzing captured network traffic and applying specific criteria to display only the packets of interest. In our case, we performed packet filtering using Wireshark by first capturing live network data and then applying built-in display filters such as IP

addresses, protocols (like TCP or HTTP), and port numbers. These filters allowed us to narrow down the large volume of captured packets and focus only on relevant communication, making it easier to study network behavior, identify specific transactions, and troubleshoot issues efficiently.

Analysing :



Packet information capture by wire shark :

Frame 111: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface eno1, id 0

Interface id: 0 (eno1)
Interface name: eno1
Encapsulation type: Ethernet (1)
Arrival Time: Mar 26, 2026 14:29:18.986086696 IST
[Time shift for this packet: 0.000000000 seconds]
Epoch Time: 1774515558.986086696 seconds
[Time delta from previous captured frame: 0.000112552 seconds]
[Time delta from previous displayed frame: 0.000112552 seconds]
[Time since reference or first frame: 0.597914308 seconds]
Frame Number: 111
Frame Length: 74 bytes (592 bits)
Capture Length: 74 bytes (592 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:tcp]

[Coloring Rule Name: TCP SYN/FIN]
[Coloring Rule String: tcp.flags & 0x02 || tcp.flags.fin == 1]
Ethernet II, Src: HewlettP_36:ba:3b (10:e7:c6:36:ba:3b), Dst: MurataMa_9f:49:5e (c4:ac:59:9f:49:5e)
Destination: MurataMa_9f:49:5e (c4:ac:59:9f:49:5e)
Address: MurataMa_9f:49:5e (c4:ac:59:9f:49:5e)
.... ..0. .... = LG bit: Globally unique address (factory default)
....0 .... = IG bit: Individual address (unicast)
Source: HewlettP_36:ba:3b (10:e7:c6:36:ba:3b)
Address: HewlettP_36:ba:3b (10:e7:c6:36:ba:3b)
.... ..0. .... = LG bit: Globally unique address (factory default)
....0 .... = IG bit: Individual address (unicast)
Type: IPv4 (0x0800)
Internet Protocol Version 4, Src: 172.180.15.59, Dst: 172.180.8.197
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
0000 00.. = Differentiated Services Codepoint: Default (0)
.... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
Total Length: 60
Identification: 0x9439 (37945)
Flags: 0x40, Don't fragment
0... .... = Reserved bit: Not set
.1.. .... = Don't fragment: Set
..0. .... = More fragments: Not set
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: TCP (6)
Header Checksum: 0x351a [validation disabled]
[Header checksum status: Unverified]
Source Address: 172.180.15.59
Destination Address: 172.180.8.197
Transmission Control Protocol, Src Port: 51500, Dst Port: 443, Seq: 0, Len: 0
Source Port: 51500
Destination Port: 443
[Stream index: 2]
[Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 0]
Sequence Number: 0 (relative sequence number)
Sequence Number (raw): 2729356135
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 0
Acknowledgment number (raw): 0
1010 .... = Header Length: 40 bytes (10)

Flags: 0x002 (SYN)
000. .... = Reserved: Not set
...0 .... = Nonce: Not set
.... 0... .... = Congestion Window Reduced (CWR): Not set
.... .0.. .... = ECN-Echo: Not set
.... ..0. .... = Urgent: Not set
....0 .... = Acknowledgment: Not set
.... 0... = Push: Not set
.... .0.. = Reset: Not set
....1. = Syn: Set
[Expert Info (Chat/Sequence): Connection establish request (SYN): server port 443]
[Connection establish request (SYN): server port 443]
[Severity level: Chat]
[Group: Sequence]
....0 = Fin: Not set
[TCP Flags: ··· ·····S·]
Window: 64240
[Calculated window size: 64240]
Checksum: 0x7197 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
Options: (20 bytes), Maximum segment size, SACK permitted, Timestamps, No-Operation (NOP), Window scale
TCP Option - Maximum segment size: 1460 bytes
Kind: Maximum Segment Size (2)
Length: 4
MSS Value: 1460
TCP Option - SACK permitted
Kind: SACK Permitted (4)
Length: 2
TCP Option - Timestamps: TSval 20429979, TSecr 0
Kind: Time Stamp Option (8)
Length: 10
Timestamp value: 20429979
Timestamp echo reply: 0
TCP Option - No-Operation (NOP)
Kind: No-Operation (1)
TCP Option - Window scale: 7 (multiply by 128)
Kind: Window Scale (3)
Length: 3
Shift count: 7
[Multiplier: 128]
[Timestamps]
[Time since first frame in this TCP stream: 0.000000000 seconds]

[Time since previous frame in this TCP stream: 0.000000000 seconds]

Analysis of above packet information :

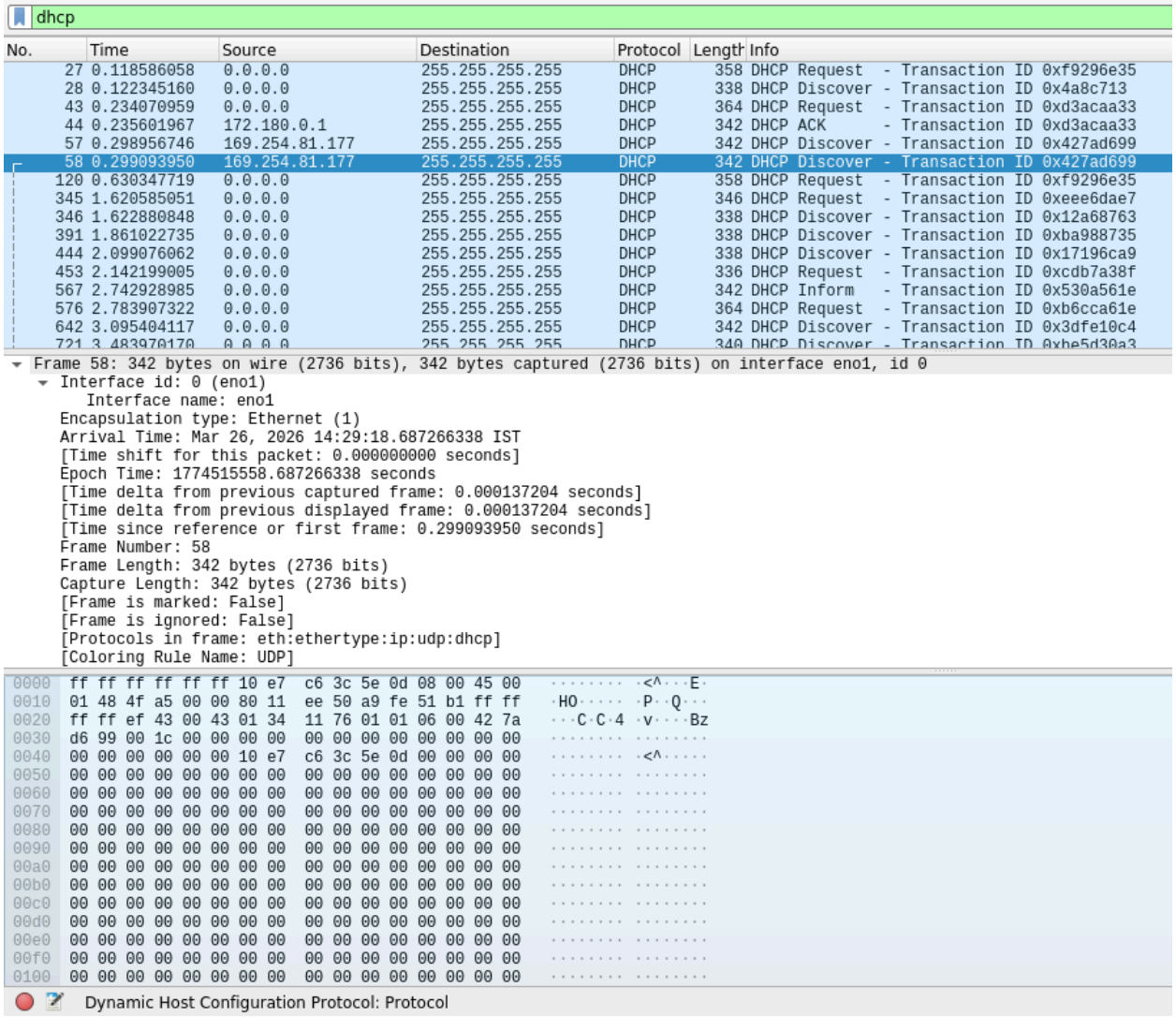
This captured packet is **Frame 111**, with a total length of **74 bytes** on the network, captured on interface `eno1`. At the **Ethernet layer**, the packet is an **Ethernet II frame**, sent from the source MAC address `10:e7:c6:36:ba:3b` (Hewlett-Packard device) to the destination MAC address `c4:ac:59:9f:49:5e` (Murata device), both being globally unique unicast addresses. The frame carries an **IPv4 packet** (Ethertype 0x0800). At the **IP layer**, the source IP is `172.180.15.59` and the destination IP is `172.180.8.197`. The IP header has a **length of 20 bytes**, **total length of 60 bytes**, and the **Don't Fragment flag** is set. The TTL (Time to Live) is 64, and the packet uses the **TCP protocol** (protocol number 6).

At the **TCP layer**, the source port is `51500` and the destination port is `443`, indicating the packet is attempting to establish an **HTTPS connection** to the server. The **sequence number** is 0, and the **SYN flag is set**, marking this as the **initial connection request** in the three-way TCP handshake. The TCP header includes **options** such as Maximum Segment Size (MSS 1460 bytes), SACK permitted, Timestamps (TSval 20429979, TSecr 0), No-Operation, and Window Scale (7, multiplier 128). The window size is 64240 bytes, allowing the receiver to specify the amount of data it can accept.

Overall, this packet represents the **first step of a TCP handshake (SYN)** from a client at `172.180.15.59` to a server at `172.180.8.197` on port 443 (HTTPS). The packet contains no payload data yet, as it is purely a connection request, and its structure confirms a properly formatted TCP/IP packet with standard options for modern TCP connections, including performance enhancements like timestamps, SACK, and window scaling. This is a normal, benign packet initiating a secure connection.

TASK-2 :

Packet capturing :



dhcp

No.	Time	Source	Destination	Protocol	Length	Info
27	0.118586058	0.0.0.0	255.255.255.255	DHCP	358	DHCP Request - Transaction ID 0xf9296e35
28	0.122345160	0.0.0.0	255.255.255.255	DHCP	338	DHCP Discover - Transaction ID 0x4a8c713
43	0.234070959	0.0.0.0	255.255.255.255	DHCP	364	DHCP Request - Transaction ID 0xd3acaa33
44	0.235601967	172.180.0.1	255.255.255.255	DHCP	342	DHCP ACK - Transaction ID 0xd3acaa33
57	0.298956746	169.254.81.177	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x427ad699
58	0.299093950	169.254.81.177	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x427ad699
120	0.630347719	0.0.0.0	255.255.255.255	DHCP	358	DHCP Request - Transaction ID 0xf9296e35
345	1.620585051	0.0.0.0	255.255.255.255	DHCP	346	DHCP Request - Transaction ID 0xeee6dae7
346	1.622880848	0.0.0.0	255.255.255.255	DHCP	338	DHCP Discover - Transaction ID 0x12a68763
391	1.861022735	0.0.0.0	255.255.255.255	DHCP	338	DHCP Discover - Transaction ID 0xba988735
444	2.099076062	0.0.0.0	255.255.255.255	DHCP	338	DHCP Discover - Transaction ID 0x17196ca9
453	2.142199005	0.0.0.0	255.255.255.255	DHCP	336	DHCP Request - Transaction ID 0xcdb7a38f
567	2.742928985	0.0.0.0	255.255.255.255	DHCP	342	DHCP Inform - Transaction ID 0x530a561e
576	2.783907322	0.0.0.0	255.255.255.255	DHCP	364	DHCP Request - Transaction ID 0xb6cca61e
642	3.095404117	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x3dfe10c4
721	3.483970170	0.0.0.0	255.255.255.255	DHCP	340	DHCP Discover - Transaction ID 0xbe5d30a3

```
Frame 58: 342 bytes on wire (2736 bits), 342 bytes captured (2736 bits) on interface eno1, id 0
  Interface id: 0 (eno1)
    Interface name: eno1
  Encapsulation type: Ethernet (1)
  Arrival Time: Mar 26, 2026 14:29:18.687266338 IST
  [Time shift for this packet: 0.000000000 seconds]
  Epoch Time: 1774515558.687266338 seconds
  [Time delta from previous captured frame: 0.000137204 seconds]
  [Time delta from previous displayed frame: 0.000137204 seconds]
  [Time since reference or first frame: 0.299093950 seconds]
  Frame Number: 58
  Frame Length: 342 bytes (2736 bits)
  Capture Length: 342 bytes (2736 bits)
  [Frame is marked: False]
  [Frame is ignored: False]
  [Protocols in frame: eth:ethertype:ip:udp:dhcp]
  [Coloring Rule Name: UDP]
```

```
0000  ff ff ff ff ff ff 10 e7  c6 3c 5e 0d 08 00 45 00   ........ .<^...E.
0010  01 48 4f a5 00 00 80 11  ee 50 a9 fe 51 b1 ff ff   .HO..... .P..Q...
0020  ff ff ef 43 00 43 01 34  11 76 01 01 06 00 42 7a   ...C.C.4 .v....Bz
0030  d6 99 00 1c 00 00 00 00  00 00 00 00 00 00 00 00   ........ ........
0040  00 00 00 00 00 00 10 e7  c6 3c 5e 0d 00 00 00 00   ........ .<^.....
0050  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   ........ ........
0060  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   ........ ........
0070  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   ........ ........
0080  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   ........ ........
0090  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   ........ ........
00a0  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   ........ ........
00b0  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   ........ ........
00c0  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   ........ ........
00d0  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   ........ ........
00e0  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   ........ ........
00f0  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   ........ ........
0100  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   ........ ........
```

Dynamic Host Configuration Protocol: Protocol

Packet information capture by wire shark :

Frame 58: 342 bytes on wire (2736 bits), 342 bytes captured (2736 bits) on interface eno1, id 0

Interface id: 0 (eno1)

Interface name: eno1

Encapsulation type: Ethernet (1)
Arrival Time: Mar 26, 2026 14:29:18.687266338 IST
[Time shift for this packet: 0.000000000 seconds]
Epoch Time: 1774515558.687266338 seconds
[Time delta from previous captured frame: 0.000137204 seconds]
[Time delta from previous displayed frame: 0.000137204 seconds]
[Time since reference or first frame: 0.299093950 seconds]
Frame Number: 58
Frame Length: 342 bytes (2736 bits)
Capture Length: 342 bytes (2736 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:udp:dhcp]
[Coloring Rule Name: UDP]
[Coloring Rule String: udp]
Ethernet II, Src: HewlettP_3c:5e:0d (10:e7:c6:3c:5e:0d), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
Destination: Broadcast (ff:ff:ff:ff:ff:ff)
Address: Broadcast (ff:ff:ff:ff:ff:ff)
.... ..1. .... = LG bit: Locally administered address (this is NOT the factory default)
....1 .... = IG bit: Group address (multicast/broadcast)
Source: HewlettP_3c:5e:0d (10:e7:c6:3c:5e:0d)
Address: HewlettP_3c:5e:0d (10:e7:c6:3c:5e:0d)
.... ..0. .... = LG bit: Globally unique address (factory default)
....0 .... = IG bit: Individual address (unicast)
Type: IPv4 (0x0800)
Internet Protocol Version 4, Src: 169.254.81.177, Dst: 255.255.255.255
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
0000 00.. = Differentiated Services Codepoint: Default (0)
.... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
Total Length: 328
Identification: 0x4fa5 (20389)
Flags: 0x00
0... .... = Reserved bit: Not set
.0.. .... = Don't fragment: Not set
..0. .... = More fragments: Not set
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 128
Protocol: UDP (17)
Header Checksum: 0xee50 [validation disabled]
[Header checksum status: Unverified]
Source Address: 169.254.81.177

Destination Address: 255.255.255.255
User Datagram Protocol, Src Port: 61251, Dst Port: 67
Source Port: 61251
Destination Port: 67
Length: 308
Checksum: 0x1176 [unverified]
[Checksum Status: Unverified]
[Stream index: 26]
[Timestamps]
[Time since first frame: 0.000000000 seconds]
[Time since previous frame: 0.000000000 seconds]
UDP payload (300 bytes)
Dynamic Host Configuration Protocol (Discover)
Message type: Boot Request (1)
Hardware type: Ethernet (0x01)
Hardware address length: 6
Hops: 0
Transaction ID: 0x427ad699
Seconds elapsed: 28
Bootp flags: 0x0000 (Unicast)
0... .... = Broadcast flag: Unicast
.000 0000 0000 0000 = Reserved flags: 0x0000
Client IP address: 0.0.0.0
Your (client) IP address: 0.0.0.0
Next server IP address: 0.0.0.0
Relay agent IP address: 0.0.0.0
Client MAC address: HewlettP_3c:5e:0d (10:e7:c6:3c:5e:0d)
Client hardware address padding: 00000000000000000000
Server host name not given
Boot file name not given
Magic cookie: DHCP
Option: (53) DHCP Message Type (Discover)
Length: 1
DHCP: Discover (1)
Option: (61) Client identifier
Length: 7
Hardware type: Ethernet (0x01)
Client MAC address: HewlettP_3c:5e:0d (10:e7:c6:3c:5e:0d)
Option: (12) Host Name
Length: 15
Host Name: DESKTOP-VL0BH1B
Option: (60) Vendor class identifier
Length: 8
Vendor class identifier: MSFT 5.0

Option: (55) Parameter Request List
Length: 14
Parameter Request List Item: (1) Subnet Mask
Parameter Request List Item: (3) Router
Parameter Request List Item: (6) Domain Name Server
Parameter Request List Item: (15) Domain Name
Parameter Request List Item: (31) Perform Router Discover
Parameter Request List Item: (33) Static Route
Parameter Request List Item: (43) Vendor-Specific Information
Parameter Request List Item: (44) NetBIOS over TCP/IP Name Server
Parameter Request List Item: (46) NetBIOS over TCP/IP Node Type
Parameter Request List Item: (47) NetBIOS over TCP/IP Scope
Parameter Request List Item: (119) Domain Search
Parameter Request List Item: (121) Classless Static Route
Parameter Request List Item: (249) Private/Classless Static Route (Microsoft)
Parameter Request List Item: (252) Private/Proxy autodiscovery
Option: (255) End
Option End: 255
Padding: 00000000

Analysis of above packet information :

Frame 58 is a **342-byte Ethernet frame** captured on interface `eno1` and represents a **DHCP Discover message**, the initial step in the DHCP process. At the **Ethernet layer**, the frame is broadcast from the client MAC address `10:e7:c6:3c:5e:0d` (Hewlett-Packard device) to the broadcast MAC `ff:ff:ff:ff:ff:ff`, indicating that it is intended for all devices on the local network. This broadcast ensures that any DHCP server on the network can receive the request. The Ethernet type field specifies **IPv4 (0x0800)**.

At the **IP layer**, the source IP is `169.254.81.177`, which is an **Automatic Private IP Addressing (APIPA)** address assigned because the client does not yet have a valid IP. The destination IP is the broadcast address `255.255.255.255`. The IP header shows a **TTL of 128**, total length of 328 bytes, and UDP as the transport protocol (protocol 17). No fragmentation is indicated.

At the **UDP layer**, the packet is sent from source port `61251` (ephemeral client port) to destination port `67`, the standard **DHCP server port**. The UDP payload carries the **DHCP Discover message**, a Boot Request (Message type 1), with a hardware type of Ethernet and a transaction ID `0x427ad699` to match the request with a future server response. The client IP, your IP, next server IP, and relay agent IP are all `0.0.0.0` because this is an initial broadcast before any assignment.

The **DHCP options** in this packet include:

- **Option 53 (DHCP Message Type)**: Discover, signaling the client is searching for available DHCP servers.
- **Option 61 (Client Identifier)**: Client MAC address.
- **Option 12 (Host Name)**: `DESKTOP-VL0BH1B`, identifying the client's machine.
- **Option 60 (Vendor Class Identifier)**: `MSFT 5.0`, indicating the client OS/vendor.
- **Option 55 (Parameter Request List)**: Requests subnet mask, router, DNS servers, domain name, and other optional parameters like NetBIOS, domain search, and classless static routes.
- **Option 255 (End)** marks the end of options.

Overall, this packet represents the **initial DHCP Discover broadcast** sent by a client attempting to obtain an IP configuration from a DHCP server. It is entirely a broadcast message with no data payload for application use, but it contains all the necessary DHCP options to request a full network configuration. The use of broadcast at both Ethernet and IP layers ensures it reaches any DHCP server on the local subnet. This is a normal, expected step in the **DORA (Discover → Offer → Request → ACK)** process.

Program 2: Constrcut a program in cryptosystem using two keys where Alice sends a message to Bob, and the same message is correctly displayed at both sender and receiver.

Code :

```
import java.security.*;
import javax.crypto.Cipher;
import java.util.Base64;
import java.util.Scanner;

public class TwoKeyCryptosystem {

	public static void main(String[] args) {
	try {
	Scanner scanner = new Scanner(System.in);
	// Step 1: Generate Bob's Key Pair
	KeyPairGenerator keyGen = KeyPairGenerator.getInstance("RSA");
	keyGen.initialize(2048);
	KeyPair bobKeyPair = keyGen.generateKeyPair();
	PublicKey bobPublicKey = bobKeyPair.getPublic();
	PrivateKey bobPrivateKey = bobKeyPair.getPrivate();

	System.out.println("Bob's Public Key (Base64): " +
Base64.getEncoder().encodeToString(bobPublicKey.getEncoded()));
```

```
        // Step 2: Take Alice's Message from User
        System.out.print("\nEnter the message Alice wants to send to Bob: ");
        String message = scanner.nextLine();
        // Encrypt the message using Bob's public key
        Cipher encryptCipher =
Cipher.getInstance("RSA/ECB/OAEPWithSHA-256AndMGF1Padding");
        encryptCipher.init(Cipher.ENCRYPT_MODE, bobPublicKey);
        byte[] encryptedMessage = encryptCipher.doFinal(message.getBytes());

        System.out.println("\nEncrypted Message (Base64): " +
Base64.getEncoder().encodeToString(encryptedMessage));
        // Step 3: Bob Decrypts the Message
        Cipher decryptCipher =
Cipher.getInstance("RSA/ECB/OAEPWithSHA-256AndMGF1Padding");
        decryptCipher.init(Cipher.DECRYPT_MODE, bobPrivateKey);
        byte[] decryptedMessage = decryptCipher.doFinal(encryptedMessage);

        System.out.println("\nDecrypted Message at Bob's Side: " + new
String(decryptedMessage));

        scanner.close();

        } catch (Exception e) {
        e.printStackTrace();
        }
        }
}
```

Output :

```
Bob's Public Key (Base64): MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAhkGT9BKIo
uc4NlvPWKku6sc5kojG8/Htfwe/VNTven3csFEMWvu+RlIeXN/NNeyO6PEnnEne4IMUwhSdVqUfPyVYL
2429OP4+MOkgiP8pwuNWBY2y/YPXAvWr4s3tnNDp4M7CPppcp+PQ51ILUj7hD0YqpasKZkqDmJ8BkYLR
hV/6R6ZYP5YblS9aLobS0rhYWL4PlTfWciAFZ/IWEmlJxx+v+mI1tl0JXKmORZQvtM/FlkZvZG93Tc1c
OykGTZ955h6NTCv5DjL3ZZ0ulo1NfBJG/I0TKZKrEJsFo+TXpnE/dY6hTXvNLlYIuk+NS9EQQE5t4iFz
vAqdFwi8n08gQIDAQAB

Enter the message Alice wants to send to Bob: hi this is vignesh

Encrypted Message (Base64): S4RqUsKMnmhT/bx0LyzsxGY4RUEcDb/x6sltFiISsnMVzs2kmc5v
CaI4o0zsgUUI1idBFYsHB7mCZsJgrBylOWCMDREWCK2OauPxsQ2LLb7nbXSOogJekArmXWkd46gpz6fC
zie0QPAIMVk49kBR2Wd8HBuiV/bc1J9QeZqA1hMuy79D0Azz/FxYeyWKOkW8CulNtvFxH7Z/BpCeuwx/
yVuCvU2vAzd72SaCfDtC9oU52thy40r69gHCfqHMDR8qwdnlwHvwYL6q9UOHQOJJUNoMd23++xmltYbo
Bbz/CwwZLutrpP2WivoYZX57O8rFyV3KqwuspT7gVG5oEabTBQ==

Decrypted Message at Bob's Side: hi this is vignesh
```